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MAST20030: Differential Equations
Tutorial Solutions: Week 10

Solution to Question 1.

(a) The steady-state solution satisfies $\dfrac{\mathrm{d}^2 u_S}{\mathrm{d}x^2} = 0$ and so $u_S(x) = cx + d$. Applying the two inhomogeneous Neumann boundary conditions gives $\partial_x u_S(x) = c$ and so $c = b$ and the steady-state solution is $u_S(x) = bx + d$ where d is arbitrary.

Now $u_T(x,t) = u(x,t) - u_S(x)$ so the IBVP for $u_T(x,t)$ is

$$\frac{\partial u_T}{\partial t} = \alpha \frac{\partial^2 u_T}{\partial x^2}, \quad \partial_x u_T(0,t) = 0, \quad \partial_x u_T(L,t) = 0 \quad \text{and} \quad u_T(x,0) = f(x) - bx - d.$$

(b) We first trial a separable solution, i.e., $u_T(x,t) = X(x)T(t)$ which leads to $X(x)\dot{T}(t) = \alpha X''(x)T(t)$. Rearranging gives

$$\frac{1}{\alpha}\frac{\dot{T}(t)}{T(t)} = \frac{X''(x)}{X(x)} \qquad \Rightarrow \qquad \frac{X''(x)}{X(x)} = -k^2 \qquad \text{and} \qquad \frac{1}{\alpha}\frac{\dot{T}(t)}{T(t)} = -k^2,$$

where k is a constant (that, in principle, may be real or imaginary as the eigenvalue $-k^2$ needs to be real due to the differential operator being self-adjoint). Substituting the separable solution into the boundary conditions gives $X'(0)T(t) = 0$ and $X'(L)T(t) = 0$ and so for non-trivial solutions we must have $X'(0) = 0$ and $X'(L) = 0$. This eigenvalue problem has solutions

$$X_n(x) = \cos k_n x \qquad \text{where} \qquad k_n = \frac{n\pi}{L}, \qquad n = 0, 1, 2, \ldots$$

Hyperbolic solutions (imaginary k) are not compatible with the boundary conditions. Solving the $T(t)$ ODE for each eigenvalue gives solutions $T_n(t) = \mathrm{e}^{-\alpha k_n^2 t}$. Taking a linear combination of all the solutions gives

$$u_T(x,t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \mathrm{e}^{-\alpha k_n^2 t} \cos k_n x.$$

(c) Applying our initial condition gives

$$u_T(x,0) = f(x) - bx - d = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos k_n x.$$

Because this is a Fourier cosine series we can just write down the formula for the coefficients,

$$a_0 = \frac{2}{L}\int_0^L \left(f(x) - bx - d\right)\,\mathrm{d}x \qquad \text{and} \qquad a_n = \frac{2}{L}\int_0^L \left(f(x) - bx - d\right)\cos k_n x\,\mathrm{d}x$$

The complete solution to (1) is therefore

$$u(x,t) = bx + d + \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \mathrm{e}^{-\alpha k_n^2 t} \cos k_n x.$$

While it looks like this solution has an undetermined constant arising from our steady state solution, d, it actually cancels out everywhere and we can write the solution as

$$u(x,t) = bx + \frac{1}{2}a_0' + \sum_{n=1}^{\infty} a_n' \mathrm{e}^{-\alpha k_n^2 t} \cos k_n x,$$

$$\text{where} \quad a_0' = \frac{2}{L}\int_0^L (f(x) - bx)\mathrm{d}x \quad \text{and} \quad a_n' = \frac{2}{L}\int_0^L (f(x) - bx)\cos k_n x \mathrm{d}x\ .$$

Solution to Question 2.

(a) Writing $u(x,t) = X(x)Y(y)$ gives

$$X''(x)Y(y) + X(x)Y''(y) = 0 \qquad \Rightarrow \qquad \frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)}.$$

Setting both sides equal to $-k^2$ gives the required ODEs:

$$X''(x) + k^2 X(x) = 0, \qquad Y''(x) - k^2 Y(y) = 0.$$

(b) The two homogeneous Dirichlet boundary conditions lead to $u(0,y) = X(0)Y(y) = 0$ and so for non-trivial solutions $X(0) = 0$, similarly $X(L) = 0$. This gives the following eigenvalues and eigenfunctions (in x)

$$k_n = \frac{\pi n}{L}, \qquad X_n(x) = \sin k_n x, \qquad n = 1, 2, 3, \ldots$$

(c) Substituting the eigenvalues into the ODE in y gives

$$Y_n''(y) - k_n^2 Y_n(y) = 0 \qquad \Rightarrow \qquad Y_n(y) = A_n \cosh k_n y + B_n \sinh k_n y.$$

(d) The separated solutions are thus $X_n(x)Y_n(y)$, for $n = 1, 2, 3, \ldots$, and we may take linear combinations in order to get a general solution that satisfies the 2D Laplace equation and the first two boundary conditions:

$$u(x,t) = \sum_{n=1}^{\infty} (A_n \cosh k_n y + B_n \sinh k_n y) \sin k_n x.$$

(e) Applying the boundary condition at $y = 0$ gives

$$u(x,0) = \sum_{n=1}^{\infty} A_n \sin k_n x = 0 \qquad \Rightarrow \qquad A_n = 0 \qquad \text{for all } n.$$

Applying the final boundary condition gives

$$u(x,L) = \sum_{n=1}^{\infty} B_n \sinh k_n L \sin k_n x = \begin{cases} x, & 0 < x < L/2 \\ L - x, & L/2 < x \leqslant L \end{cases}.$$

Now if we let $b_n = B_n \sinh k_n L$ then this is a Fourier sine series and we have already calculated the coefficients in class; $b_{2m-1} = -\dfrac{4L(-1)^m}{\pi^2(2m-1)^2}$ and $b_{2m} = 0$. The complete solution is therefore

$$u(x,y) = -\frac{4L}{\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \frac{\sinh k_{2m-1} y}{\sinh k_{2m-1} L} \sin k_{2m-1} x$$